

Chapter C1: Air and water

Overview

The quality of our air and water is a major world concern. Chemists monitor our air and water, and work to minimise the impact of human activities on their quality.

In Topic C1.1, the context of changes in the Earth's atmosphere is used to explore the particle model and its limitations when explaining changes of state, and the principles of balancing equations for combustion reactions. As a development of ideas about burning fuels, Topic C1.2 considers bonding in small molecules and temperature changes in chemical reactions.

Topic C1.3 explores the evidence for climate change, asking why it might be occurring and how serious a threat it is. Learners consider environmental and health consequences of some air pollutants and climate change, and learn how scientists are helping to provide options for improving air quality and combatting global warming.

Finally, Topic C1.4 explores the need for increasing the amount of potable water worldwide, and techniques for obtaining potable water from ground, waste and salt water.

Learning about air and water before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- be able to explain the properties of the different states of matter (solid, liquid and gas) in terms of the particle model, including gas pressure
- appreciate the differences between atoms, elements and compounds
- be familiar with the use of chemical symbols and formulae for elements and compounds
- know about conservation of mass, changes of state and chemical reactions
- be able to explain changes of state in terms of the particle model
- know that there are energy changes on changes of state (qualitative)
- know about exothermic and endothermic chemical reactions (qualitative)
- understand the carbon cycle
- know the composition of the Earth's atmosphere today
- know about the production of carbon dioxide by human activity and its impact on climate.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about air and water at GCSE (9–1)
